

# A novel approach to characterizing the visual dimensions of mental imagery vividness using Gaussian process regression

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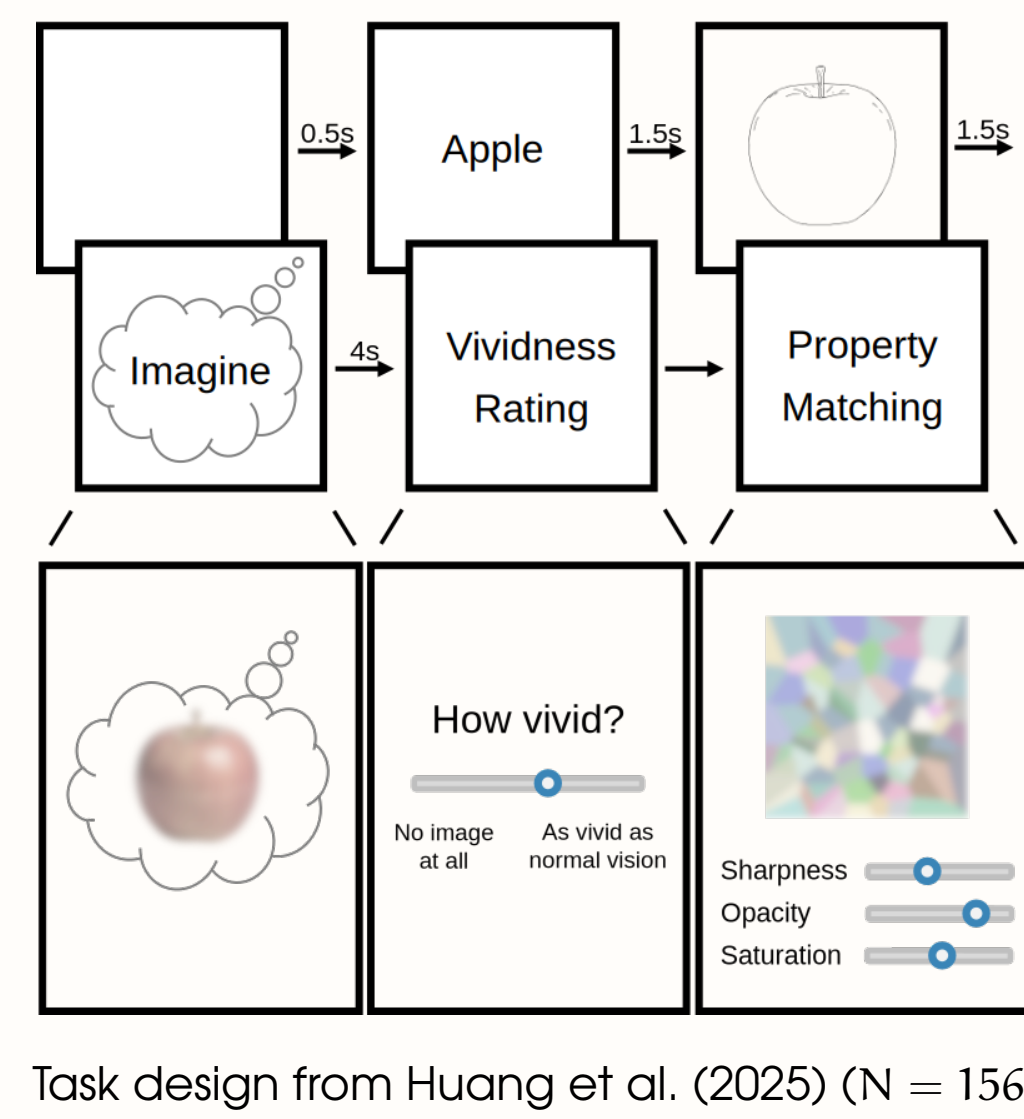
## (1) How to characterise vividness?

Huang et al. (2025) used linear regression to reveal evidence of common structure in 'vividness' & large individual differences.

### Current aims

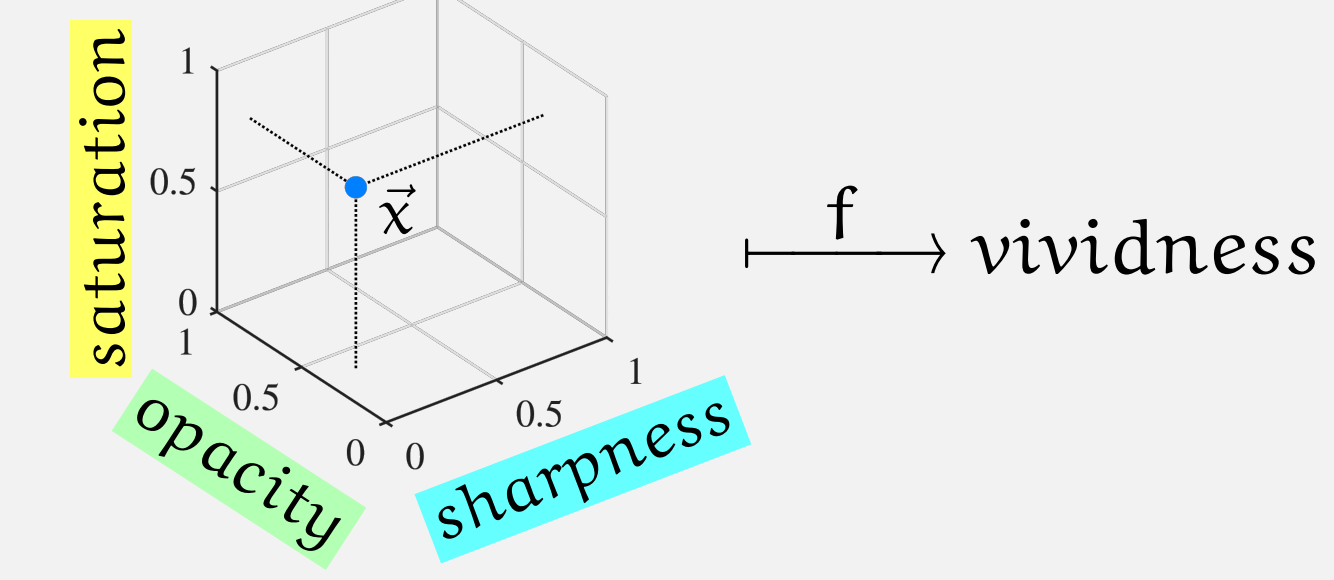
- (1) Move beyond the parametric assumptions of linear models
- (2) Characterise individual differences in a principled manner

## (2) Data and problem formulation



Vividness ratings (1 – 5 continuous scale)  
Perceptual features (0 – 1 continuous scale): sharpness, opacity, saturation

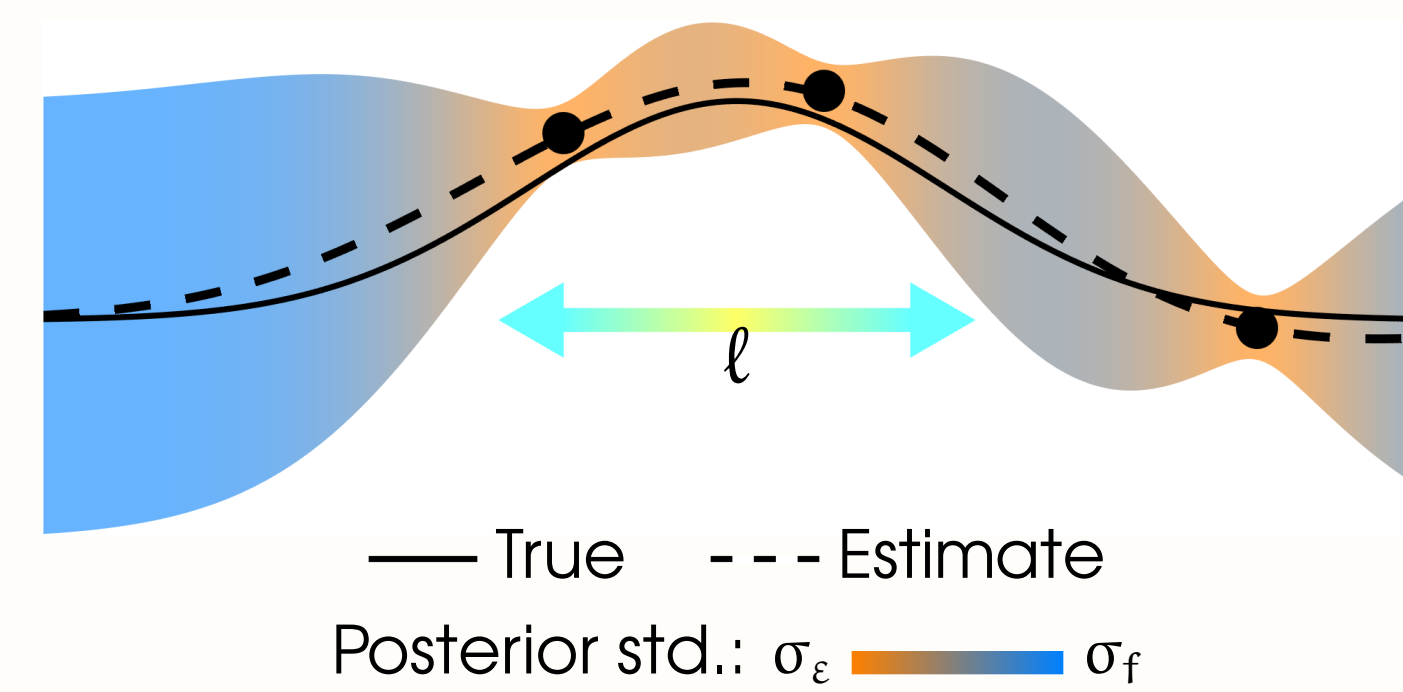
What is the function  $f(\cdot)$  that maps these features onto vividness?



## (3) What are Gaussian processes?

**Setup:** N observations  $y_i$  of  $f(\cdot)$  at input locations  $\vec{x}_i$  such that  $y_i = f(\vec{x}_i) + \varepsilon_i$

**Core idea:** Finite sets of evaluations of  $f(\vec{x})$  are jointly Gaussian with the covariance structure determined by inputs  $\vec{x}$  (Rasmussen and Williams 2005)



$$\text{Cov}[y_i, y_j] = k(\vec{x}_i, \vec{x}_j) + \sigma_\varepsilon^2 \delta_{ij}$$

$$= \sigma_f^2 \exp \left( -\frac{1}{2} \left( \frac{(\vec{x}_i^{\text{sha}} - \vec{x}_j^{\text{sha}})^2}{\ell_{\text{sharpness}}^2} + \frac{(\vec{x}_i^{\text{opa}} - \vec{x}_j^{\text{opa}})^2}{\ell_{\text{opacity}}^2} + \frac{(\vec{x}_i^{\text{sat}} - \vec{x}_j^{\text{sat}})^2}{\ell_{\text{saturation}}^2} \right) \right) + \sigma_\varepsilon^2 \delta_{ij}$$

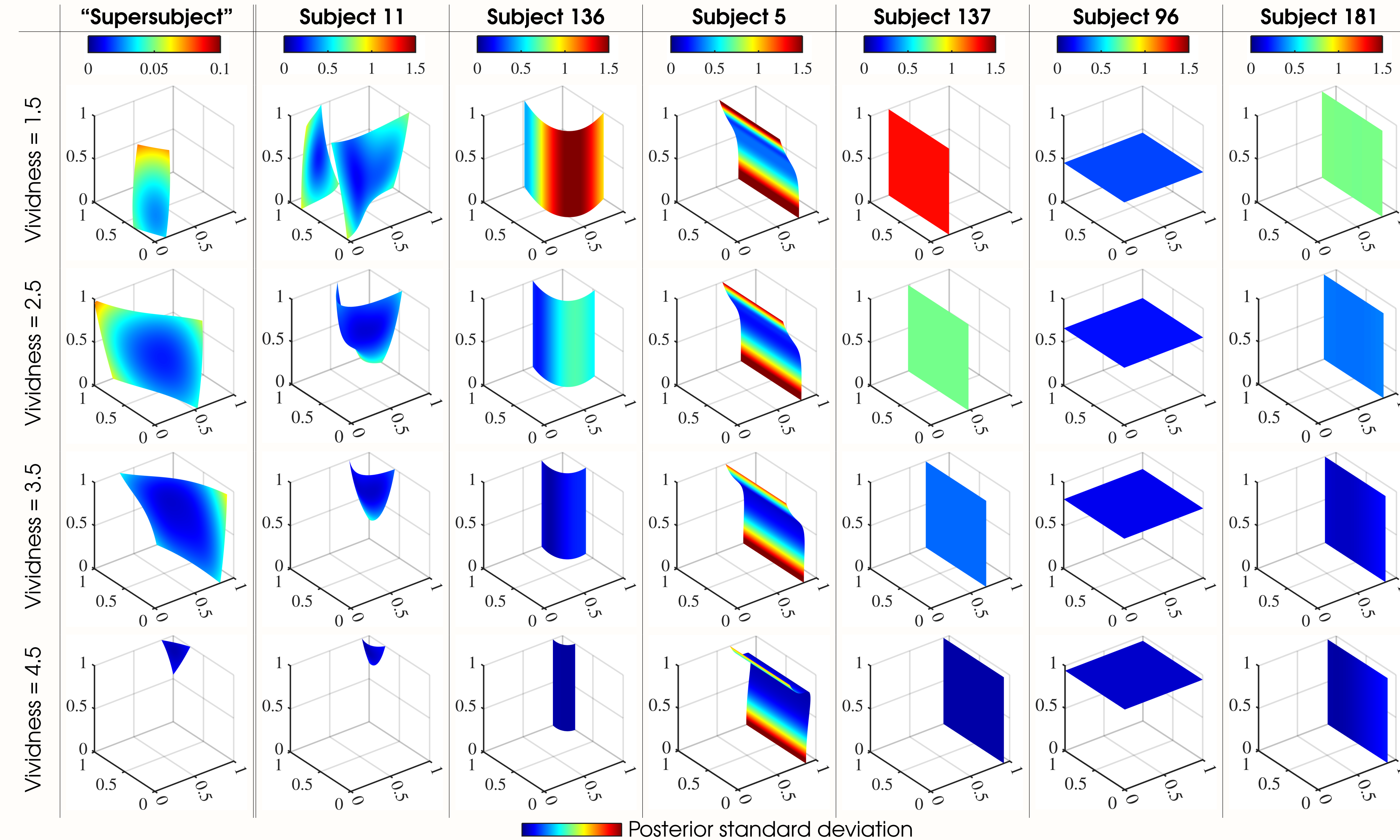
where  $\delta_{ij}$  is the impulse function that takes the value 1 only when  $i = j$  and 0 otherwise.

### Hyperparameters:

- Kernel function:** Covariance between noiseless function evaluations
- Noise standard deviation:** Uncertainty on measurements  $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$
- Signal standard deviation:** Uncertainty on unobserved locations
- Length scales:** Smoothness over variables

Model fits are obtained using MATLAB Statistics and Machine Learning Toolbox (2026).

## (4) Gaussian processes reveal nonlinearities and clusters of individuals

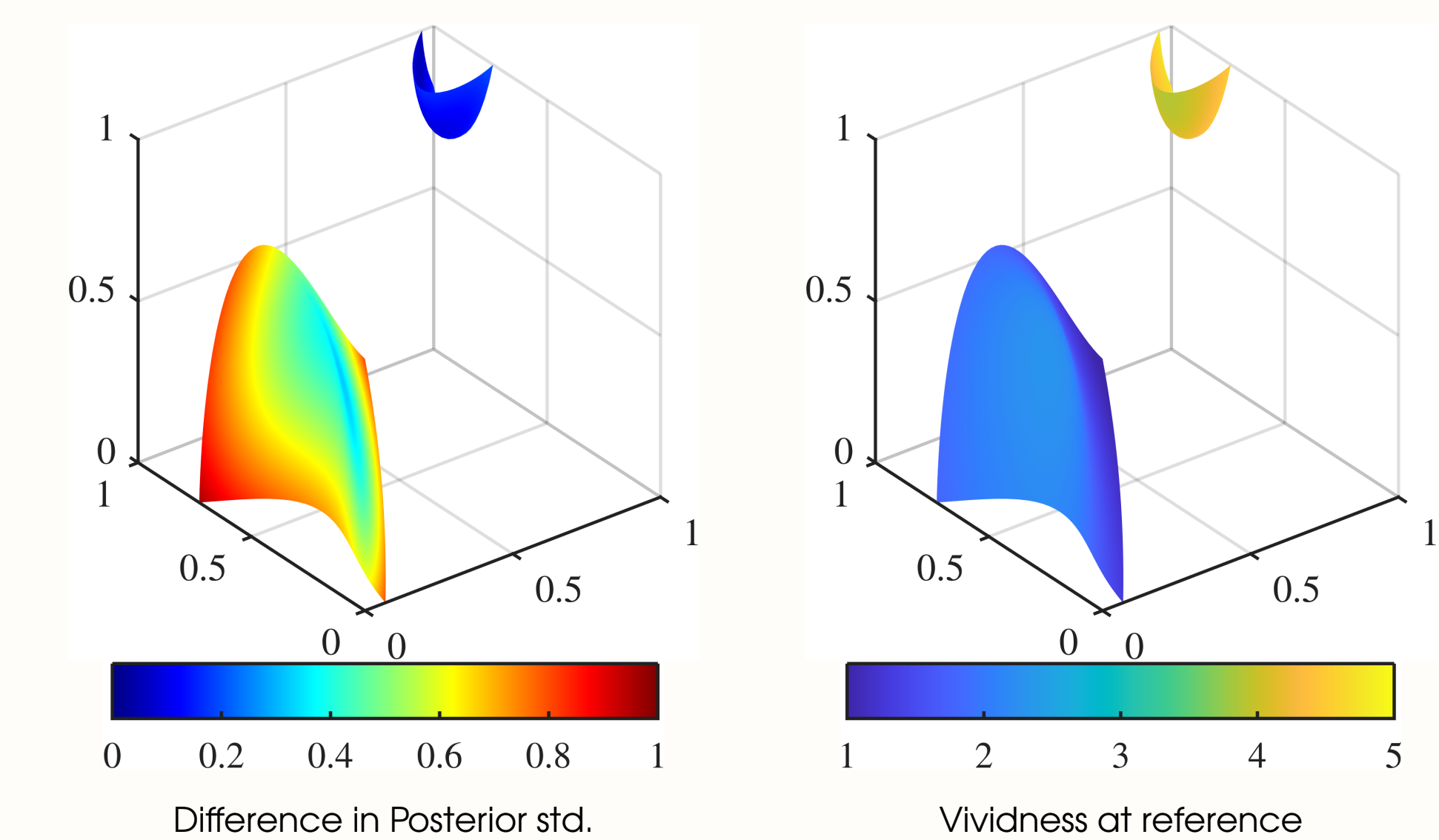


### Equal vividness manifolds

where different features correspond to the same vividness rating

### Difference manifolds

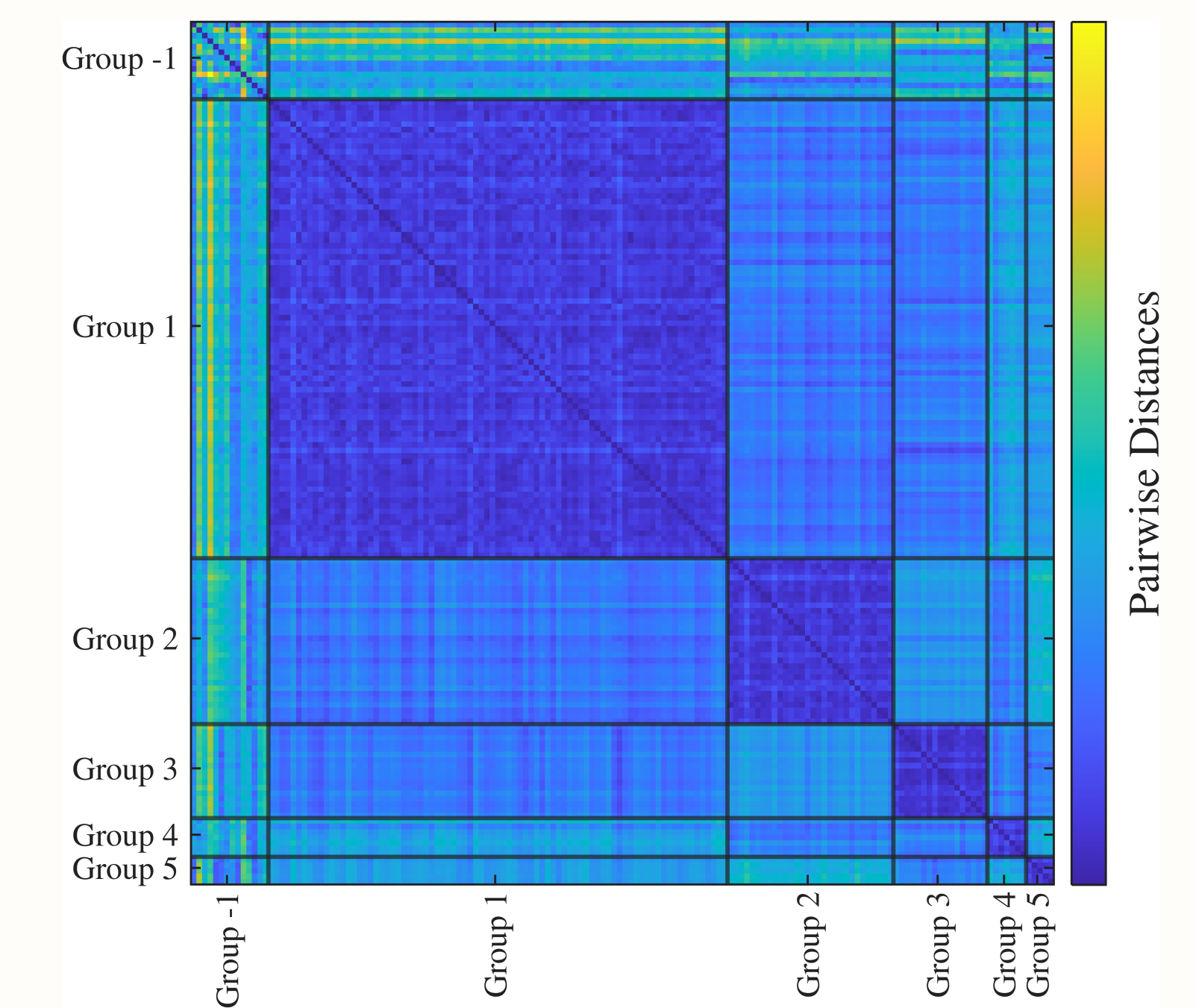
where the difference  $\Delta$  between two profiles is constant



Figures:  $\Delta(\text{Subject 11} - \text{Supersubject}) = 0$

**Non-planar manifolds imply that nonlinear relationships can be captured using Gaussian processes!**

### Hyperparameter-based clustering



Clustered using DBSCAN (Ester et al. 1996) and Satopaa et al. (2011) for knee detection.

**Groups differ mostly in the way they are sensitive to the three perceptual features!**

## (5) Take-home messages

### Vividness ratings...

- (1) can show nonlinear dependence on perceptual features, as seen through non-planar equal vividness manifolds over the feature space!
- (2) show individual differences in their dependence on these features, agreeing with Huang et al. (2025), ...yet there are trends in these differences!

### Why use Gaussian processes?

Non-parametric regression method not restricted by functional form  
Inherently generative model  
Model confidence/prediction intervals come naturally  
Not a black box model: hyperparameters are interpretable

### Gaussian processes can help answer...

How do individuals differ from the general population? From each other?  
How to choose input values to elicit a specific response for any multivariate nonlinear behavioral output, without parametric assumptions?  
... and in a way that agrees with the whole population?

### References

- Huang, Shen, Olsson, Garcia, Dijkstra, Peters<sup>†</sup>, Morales<sup>†</sup> (2025). "What makes mental images vivid? Sharpness as the key visual dimension". en. In: *Journal of vision*.  
Rasmussen, Williams (Nov. 2005). *Gaussian processes for machine learning*. en. Adaptive Computation and Machine Learning series. London, England: MIT Press.  
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